ALLEN CAREER INSTITUTE KOTA (RAJASTHAN)

DISTANCE LEARNING PROGRAMME

(Academic Session : 2024 - 2025)

JEE (Advanced)
TEST # 05
01-09-2024

JEE(Main + Advanced) : LEADER TEST SERIES / JOINT PACKAGE COURSE

Test Type : Unit Test # 04

ANSWER KEY

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4		
	A.	A	A	B	A		
SECTION-I (ii)	Q.	5	6	7	8	9	10
	A.	B,C	A,B,C,D	A,D	A,B	A,C,D	A,B
SECTION-I (iii)	Q.	11	12	13	14		
	A.	A	B	C	A		
SECTION-III	Q.	1	2	3	4	5	6
	A.	2	2	1	4	2	4
SECTION-IV	Q.	1	2				
	A.	A->QR,B->P,C->T,D->S	A->S,B->PT,C->QT,D->R				

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4		
	A.	C	B	B	B		
SECTION-I (ii)	Q.	5	6	7	8	9	10
	A.	B,D	A,C,D	A,B,C	A,B,C	B,D	B,C
SECTION-I (iii)	Q.	11	12	13	14		
	A.	B	D	A	C		
SECTION-III	Q.	1	2	3	4	5	6
	A.	8	1	4	6	6	3
SECTION-IV	Q.	1	2				
	A.	A->PRT,B->PST,C->PRT,D->QR	A->PQS,B->T,C->QR,D->R				

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4		
	A.	C	B	D	B		
SECTION-I (ii)	Q.	5	6	7	8	9	10
	A.	A,B,C,D	A,B	A,B,D	A,B,C	A,C	B,C
SECTION-I (iii)	Q.	11	12	13	14		
	A.	B	C	B	C		
SECTION-III	Q.	1	2	3	4	5	6
	A.	4	5	5	0	8	4
SECTION-IV	Q.	1	2				
	A.	A->S,B->R,C->T,D->Q	A->P,B->S,C->R,D->Q				

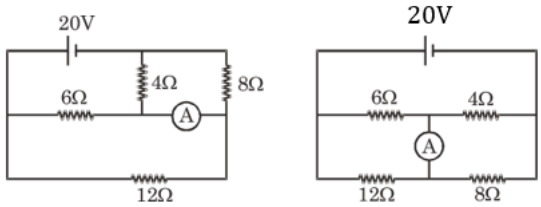
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HINT – SHEET

PART-1 : PHYSICS

SECTION-I (i)

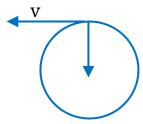
1. Ans (A)



$\frac{6}{12} = \frac{4}{8}$

So, it will act as wheatstone bridge, current through ammeter ⟶ zero

2. Ans (A)



$T + mg = \frac{MV^2}{R}$

$T = 16 - 10$

$T = 6W$

3. Ans (B)

$\theta = \frac{1}{2} \times 4 \times 3^2 = 18$

$n = \frac{18}{2\pi} = \frac{9}{3.14} = 2.9$

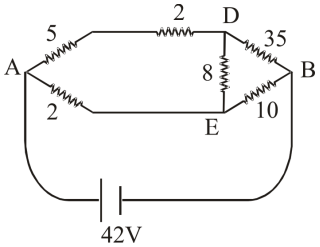
4. Ans (A)

area under graph will be equal to angular displacement $\omega_{av} = \frac{\theta}{t}$

PART-1 : PHYSICS

SECTION-I (ii)

5. Ans (B,C)



It's a balanced WSB so $P_{8\Omega} = 0$ &

$P_{5\Omega} = \left(\frac{42}{42}\right)^2 \cdot 5 = 5W$

6. Ans (A,B,C,D)

$\frac{1}{20} = \frac{1}{20+R} + \frac{1}{25}$

$\therefore \; R = 80\,\Omega$

Bridge is balanced.

$\therefore$ No current flows.

$P_R = I^2R = I^2 \times 80$

$P_S = (4I)^2 \times 5 = I^2 \times 80$

$P_{20} = I^2 \times 20$ (upper line)

$= (4I)^2 \times 20$ (lower line)

9. Ans (A,C,D)

As i changes (in same resistor) ; n remain constant

by $i = JA$; J become double

by $i = nAeV_d$; V_d become double

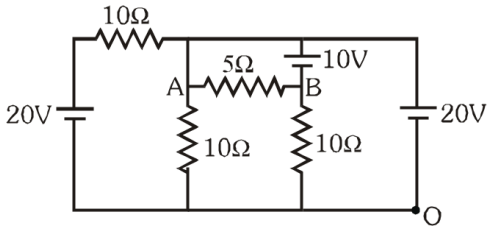
10. Ans (A,B)

If speed is less tendency of sliding is downward and friction up the plane for high speeds friction is down the plane.

PART-1 : PHYSICS

SECTION-III

1. Ans (2)



Let potential of point O is 0 volt.

Therefore $V_A = 20$ & $V_B = 10$ volt

$i = \frac{20 - 10}{5} = 2A$

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2. **Ans (2)**

Resistance per unit length

$= \frac{R}{\ell} = \frac{4.5}{3} = 1.5\ \Omega$ / meter

Now potential drop across AC should equal to 3V

and current in AB

= 5 Volt/(0.5 + 4.5 Ω) = 1 AmP

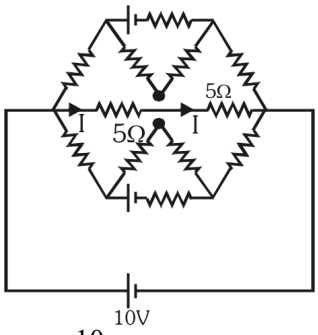
Now, 3 Volt = I × R

$3 = 1 \times (1.5) \times \ell$

$\ell = \frac{3}{1.5} = 2$ meter

3. **Ans (1)**

By symmetry circuit can be reduced to



$I = \frac{10}{5+5} = 1A$

4. **Ans (4)**

$\omega_0 = 12$ rad/sec

$\Delta\theta = 80$ radian

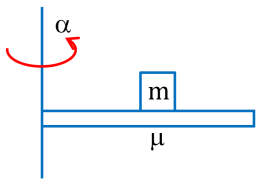
$\omega = 28$ radian/sec

$\omega^2 = \omega_0^2 + 2\alpha\Delta\theta$ ...(i)

$\omega = \omega_0 + \alpha t$ ...(ii)

$\left(\frac{\omega_0 + \omega}{2}\right) t = 80$

5. **Ans (2)**



$\mu mg = m\sqrt{a_t^2 + a_n^2}$

$r\omega^2 = (\alpha t)^2 = x^2$

$(0.5 \times 10)^2 = x^4 + 9$

$x^4 = 16$

$x = 2$

6. **Ans (4)**

$\theta = \omega_0 t + \frac{1}{2}\alpha t^2$

Given : $\omega_0 = 1$ rad/s

$\alpha = 1.5$ rad/s^2

t = 2 sec

$\omega = \omega_0 + \alpha t$

$\omega = 1 + 1.5 \times 2$

= 4 rad/s

PART-1 : PHYSICS

SECTION-IV

1. **Ans (A->QR,B->P,C->T,D->S)**

v =2t

$a_t = \frac{dv}{dt} = 2$

$a_c = \frac{v^2}{R} = 2t^2$

$\alpha = \frac{a_t}{R} = 1$

$\theta = \frac{1}{2}\alpha t^2$

Angle between acceleration and velocity

$= \tan^{-1}\left(\frac{a_c}{a_t}\right)$

2. **Ans (A->S,B->PT,C->QT,D->R)**

(A) $\text{Power} = \frac{V^2_{\text{across bulb}}}{R_{\text{bulb}}}$

Since voltage across B_4 is highest, it will glow brightest

(B) $\text{Power} = i^2_{\text{bulb}} R_{\text{bulb}}$

Since i will highest in B_1, B_1 will glow brightest.

Also, current in B_2, B_3 & B_4 are same. So, they will glow at same power.

(C) B_2 will glow brightest, since voltage across it is maximum & B_1, B_3, B_4 will glow at same power since current is same.

(D) Current is highest in B_3, so it will glow brightest.

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PART-2 : CHEMISTRY

SECTION-I (i)

1. Ans (C)

If reaction is balanced

$2MnO_4^- + 10I^- + 16H^+ \longrightarrow 2Mn^{2+} + 5I_2 + 8H_2O$

PART-2 : CHEMISTRY

SECTION-I (ii)

6. Ans (A,C,D)

$\overset{+7}{K}\,\overset{+7}{Mn}\,O_4 \xrightarrow{H^+} Mn^{+2} - 5e^-$

(i) $Fe^{+2} \longrightarrow Fe^{+3} + e$

(ii) $Fe^{+2} + C_2^{+3}O_4^{-2} \longrightarrow 2\overset{+4}{C}\,O_2 + Fe^{+3} + 3e$

(iii) $H_2\overset{+3}{C}_2O_4 \longrightarrow 2\overset{+4}{C}\,O_2 + 2e^-$

(A) 5 mole of $FeSO_4$ can be oxidised

(C) 5/3 mole of FeC_2O_4 can be oxidised

(D) 2.5 mole of $H_2C_2O_4$ can be oxidised

PART-2 : CHEMISTRY

SECTION-I (iii)

11. Ans (B)

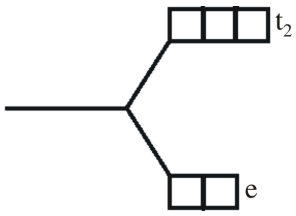
d_{xz}^2 , d_{yz}^2 , $d_{z^2}^2$, d_{xy}^2

$CFSE = 4(-0.51\Delta_0) + 2(-0.43\Delta_0)$

$+ 2(0.23\ \Delta_0) + 0(1.23\ \Delta_0)$

$= -2.44\ \Delta_0$

12. Ans (D)



de–	CFSE
d^1	$-0.6\Delta_t$
d^2	$-1.2\Delta_t$
d^3	$-0.8\Delta_t$
d^4	$-0.4\Delta_t$
d^5	-0.0
d^6	$-0.6\Delta_t$
d^7	$-1.2\Delta_t$
d^8	$-0.8\Delta_t$
d^9	$-0.4\Delta_t$
d^{10}	0.0

13. Ans (A)

(A) is correct. In the Van–Arkel method zironium is purified by vapour phase refining.

$\underset{\text{impure}}{Zr} + 2I_2 \xrightarrow{600C} ZrI_4 \xrightarrow{1800C} \underset{\text{pure}}{Zr} + 2I_2$

14. Ans (C)

(C) is correct.

Fe is not extracted by hydrometallurgy. It is extracted by pyrometallurgy.

(A) is wrong. Al is extracted by hydrometallurgy.

(i) Baeyer's process conversion of bauxite to $NaAlO_2$.

(ii) Conversion of AlO_2^- to $Al(OH)_3^-$ by hydrolysis.

(iii) $Al(OH)_3 \xrightarrow{heat} Al_2O_3$

(iv) Electrolysis of $Al_2O_3 \rightarrow Al$

(B) is wrong as Na is extracted from molten NaCl.

(D) is wrong as Ag is extracted by cyanide process which is hydrometallurgy.

PART-2 : CHEMISTRY

SECTION-III

1. Ans (8)

Let milli mole of $H_2C_2O_4$ & $K_2C_2O_4$ are x + y respectively

Number of milli equivalent of $H_2C_2O_4$ + Number of milli equivalent of $K_2C_2O_4$ = Number of milli equivalent of $KMnO_4$

$2 \times x + 2y = 0.2 \times 25$

$x + y = 2.5$ .....(1)

Number of milli equivalent of $H_2C_2O_4$ = No. of milli equivalent of NaOH

$2x = 0.2 \times 20 = 4$

$x = 2$ .....(2)

$\therefore y = 0.5$

$\therefore$ mole % of $H_2C_2O_4 = \frac{2}{2.5} \times 100 = 80\%$

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3. **Ans (4)**

$\Delta_t = \frac{4}{9}\Delta_0$

5. **Ans (6)**

Oxidation state(+1) present in any metal of following ores are :-

$\overset{(+1)\,(+2)}{CuFeS_2}$ (copper pyrite), $\overset{(+1)}{Cu_2S}$ (Chalcocite)

$\overset{(+1)}{NaNO_3}$ (Chile saltpeter); $\overset{(+1)}{Na_3AlF_6}$(Cryolite)

$\overset{(+1)}{Cu_2O}$ (Cuprite) ; Sylvine $\overset{(+1)}{(K_4)}$

PART-2 : CHEMISTRY

SECTION-IV

2. **Ans (A->PQS,B->T,C->QR,D->R)**

(P)	$FeCO_3$	(Siderite)
(Q)	$CuCO_3.Cu(OH)_2$	(Malachite)
(R)	$Al2O_3.2H_2O$, $AlOx(OH)_{3-2x}$	(Bauxite)
(S)	$ZnCO_3$	(Calamine)
(T)	Ag_2S	(Argentite)

PART-3 : MATHEMATICS

SECTION-I (i)

1. **Ans (C)**

$\cos(\theta_1-\theta_2)+\cos(\theta_2-\theta_3)+\cos(\theta_3-\theta_1)=-\frac{3}{2}$

$\Rightarrow 3+2\cos(\theta_1-\theta_2)+2\cos(\theta_2-\theta_3)+2\cos(\theta_3-\theta_1)=0$

$\Rightarrow (\sin\theta_1+\sin\theta_2+\sin\theta_3)^2+(\cos\theta_1+\cos\theta_2+\cos\theta_3)^2=0$

$\Rightarrow \sin\theta_1+\sin\theta_2+\sin\theta_3=0 \,\&\, \cos\theta_1+\cos\theta_2+\cos\theta_3=0$

2. **Ans (B)**

$\sin x = -\frac{1}{2}$ and $\tan x = \frac{1}{\sqrt{3}}$

3. **Ans (D)**

$\frac{dy}{dx} = \frac{x^2}{4+x^2}$ and $\cot\left(\frac{y-x}{2}\right) = \frac{x}{2}$

$I = \int \frac{\left(\frac{x^2}{4+x^2}\right)dx}{\left(\frac{x^2-4}{x^2+4}\right)\sin x + \frac{4x}{x^2+4}\cos x}$

$= \int \frac{dy}{\cos(y-x)\sin x + \sin(y-x)\cos x} = \int \frac{dy}{\sin y}$

4. **Ans (B)**

$\int \tan\left(\frac{2\pi}{3}+\frac{x}{2}\right).\tan\left(\frac{\pi}{6}-\frac{x}{4}\right).\tan\left(\frac{\pi}{6}-\frac{x}{4}\right)dx$

$= \int \tan\left(\frac{2\pi}{3}+\frac{x}{2}\right)+\tan\left(\frac{\pi}{6}-\frac{x}{4}\right)+\tan\left(\frac{\pi}{6}-\frac{x}{4}\right)dx$

$= 2\ell n\sec\left(\frac{2\pi}{3}+\frac{x}{2}\right)-4\ell n\sec\left(\frac{\pi}{6}-\frac{x}{4}\right)-4\ell n\sec\left(\frac{\pi}{6}-\frac{x}{4}\right)$

PART-3 : MATHEMATICS

SECTION-I (ii)

5. **Ans (A,B,C,D)**

$\sec^2\alpha + 6\cos^2\alpha = p$ ...(1)

$\tan^2\alpha - 5\sin\alpha = p - 8$ ...(2)

(1) − (2) $\Rightarrow 6 - 6\sin^2\alpha - 5\sin\alpha = 7$

$\Rightarrow (2\sin a - 1)(3\sin a - 1) = 0$

$\Rightarrow \sin\alpha = \frac{1}{2}$ or $\sin\alpha = \frac{1}{3}$

6. **Ans (A,B)**

$\sin^2 x + \sin x = p + 2$

$\Rightarrow \left(\sin x + \frac{1}{2}\right)^2 - \frac{1}{4} = p + 2$

$\therefore \frac{-1}{4} \leqslant p + 2 \leqslant 2$

$\Rightarrow \frac{-9}{4} \leqslant p \leqslant 0$

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7. **Ans (A,B,D)**

For $x \in (0,\pi) - \left\{\frac{\pi}{2}\right\}$

$x \in (0,\pi) - \left\{\frac{\pi}{2}\right\}$

$\sqrt{3}\cos x + \sin x = 2\sin 2x$

$\frac{\sqrt{3}}{2}\cos x + \frac{1}{2}\sin x = \sin 2x$

$\sin x \cos\frac{\pi}{3} + \cos x \sin\frac{\pi}{3} = \sin 2x$

$\sin\left(x + \frac{\pi}{3}\right) = \sin 2x$

$x + \frac{\pi}{3} = n\pi + (-1)^n.2x$

When n is even When n is odd

$x + \frac{\pi}{3} = n\pi + 2x$ $x + \frac{\pi}{3} = n\pi - 2x$

$x = -n\pi + \frac{\pi}{3}$ $x = \frac{n\pi}{3} - \frac{\pi}{4}$

$n = 0$ $n = 1$ $n = 3$

$x = \frac{\pi}{3}$ $x = \frac{2\pi}{9}$ $x = \frac{8\pi}{9}$

8. **Ans (A,B,C)**

$a\cos x - \cos 2x = 2a - 7$

$(4\cos x - a)^2 = (a-8)^2$

$4\cos x = 8$ (rejected) or $\cos x = \frac{a}{2} - 2 \in [-1,1]$

$\Rightarrow a \in [2,6]$

9. **Ans (A,C)**

$$\int \frac{\cos^3 x(1+\cos^2 x)}{\sin^2 x(2-\sin^2 x)}dx$$

$$= \int \frac{(1-\sin^2 x)(2-\sin^2 x)}{\sin^2 x(2-\sin^2 x)}\cos x\,dx$$

put $\sin x = t$

$$\Rightarrow F(x) = \int \frac{1-t^2}{t^2}dt = -\frac{1}{\sin x} - \sin x + C$$

$\because F\left(\frac{\pi}{2}\right) = 0 \Rightarrow C = 2$

$$\Rightarrow F(x) = 2 - \left(\sin x + \frac{1}{\sin x}\right)$$

10. **Ans (B,C)**

$$\int \frac{\left(1-\frac{1}{x^2}\right)dx}{\sqrt{x+1+\frac{1}{x}} + \sqrt{x-1+\frac{1}{x}}}$$

put $x + \frac{1}{x} = t$

$$\Rightarrow \int \frac{dt}{\sqrt{t+1}+\sqrt{t-1}}$$

$$\Rightarrow \frac{1}{2}\int \left(\sqrt{t+1} - \sqrt{t-1}\right)dt$$

$$\Rightarrow \frac{(t+1)^{3/2} - (t-1)^{3/2}}{3} + C \Rightarrow f(x) = x + \frac{1}{x} - 1$$

PART-3 : MATHEMATICS

SECTION-I (iii)

11. **Ans (B)**

$4(\sin^4\theta + \cos^4\theta) = 1 - 4x^2 - 4x$

$\Rightarrow 4(1 - 2\sin^2\theta\cos^2\theta)$

$\Rightarrow 4\left(1 - \frac{1}{2}\sin^2 2\theta\right) = -\left(4x^2 + 4x + 1\right) + 2$

$\Rightarrow 4\left(1 - \frac{1}{2}\sin^2 2\theta\right) = 2 - (2x+1)^2$

Now $4\left(1 - \frac{1}{2}\sin^2 2\theta\right) \in [2,4]$

and $2 - (2x+1)^2 \in (-\infty, 2]$

or $x = -\frac{1}{2}$, $\sin^2 2\theta = 1$

or $2\theta = n\pi \pm \frac{\pi}{2}$

or $\theta = \frac{n\pi}{2} \pm \frac{\pi}{4}$

12. **Ans (C)**

$4(\sin^4\theta + \cos^4\theta) = 1 - 4x^2 - 4x$

$\Rightarrow 4(1 - 2\sin^2\theta\cos^2\theta)$

$\Rightarrow 4\left(1 - \frac{1}{2}\sin^2 2\theta\right) = -\left(4x^2 + 4x + 1\right) + 2$

$\Rightarrow 4\left(1 - \frac{1}{2}\sin^2 2\theta\right) = 2 - (2x+1)^2$

Now $4\left(1 - \frac{1}{2}\sin^2 2\theta\right) \in [2,4]$

and $2 - (2x+1)^2 \in (-\infty, 2]$

or $x = -\frac{1}{2}$, $\sin^2 2\theta = 1$

or $2\theta = n\pi \pm \frac{\pi}{2}$

or $\theta = \frac{n\pi}{2} \pm \frac{\pi}{4}$

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13. **Ans (B)**

$$\int x^2\left(\frac{x\sec^2x+\tan x}{(x\tan x+1)^2}\right)dx$$

$$x^2.\left(\frac{-1}{x\tan x+1}\right)+\int\frac{2x}{x\tan x+1}dx$$

$$2\ell n|x\sin x+\cos x|-\frac{x^2}{x\tan x+1}+C$$

$\Rightarrow$ $f(x) = x\sin x + \cos x$

$g(x) = x\tan x + 1$

$f(-x) = f(x) \Rightarrow$ even function

14. **Ans (C)**

$\Rightarrow$ $f(x) = x\sin x + \cos x$

$g(x) = x\tan x + 1$

$$\lim_{x\to0}\frac{x\tan x}{x\sin x+\cos x-1}$$

$$\lim_{x\to0}\frac{x\tan x}{x\sin x-2\sin^2\frac{x}{2}}$$

$$\lim_{x\to0}\frac{\frac{x\tan x}{x^2}}{\frac{x\sin x}{x^2}-\frac{2\sin^2\frac{x}{2}}{x^2}}$$

$$=\frac{1}{1-2.\frac{1}{4}}=2$$

PART-3 : MATHEMATICS

SECTION-III

1. **Ans (4)**

$$x=\sum_{r=0}^{179}(2r+1)\sin(2r+1)^\circ$$

$x = \sin1^\circ + 3\sin3^\circ + 5\sin5^\circ + ...+359\sin359^\circ$

$\Rightarrow$ $x = 180(\sin1^\circ + \sin3^\circ + ... \sin89^\circ)$

$-540(\sin1^\circ + \sin3^\circ .... + \sin89^\circ)$

$\Rightarrow x = -360^\circ(\sin1^\circ + \sin3^\circ + ... + \sin89^\circ)$

$$\Rightarrow x=\frac{-360\left(\sin1^\circ+\frac{(45^\circ-1^\circ)}{2}.2\right).\sin\left(\frac{45^\circ}{2}.2\right)}{\sin\left(\frac{2^\circ}{2}\right)}$$

$$x=-\frac{360}{\sin1^\circ}.\frac{1}{2}$$

$$\therefore\ \frac{|x|\sin1^\circ}{45}=4$$

2. **Ans (5)**

Let $\theta=\frac{\pi}{16}\Rightarrow 8\theta=\frac{\pi}{2}$

$\tan^2\theta + \tan^2 2\theta + \tan^2 3\theta + \tan^2 4\theta + \tan^2 5\theta + \tan^2 6\theta + \tan^2 7\theta$

$= (\tan^2\theta + \cot^2\theta) + (\tan^2 2\theta + \cot^2 2\theta) + (\tan^2 3\theta + \cot^2 3\theta) + \tan^2 4\theta$ ...(1)

$\Rightarrow \tan^2\theta + \cot^2\theta = (\tan\theta + \cot\theta)^2 - 2$

$= (2\,\text{cosec}\,2\theta)^2 - 2$

$$=\frac{8}{1-\cos4\theta}-2$$

$= 14 + 8\sqrt{2}$

similarly $\tan^2 2\theta + \cot^2 2\theta = 6$

$\tan^2 3\theta + \cot^2 3\theta = 14 - 8\sqrt{2}$

$\tan^2 4\theta = 1$

$\Rightarrow$ Required sum is 35

3. **Ans (5)**

$$y=5\frac{\sin^2x}{1-\sin^2x}+125\frac{\cos^2x}{1-\cos^2x}+7$$

$= 5\tan^2x + 125\cot^2x + 7$

$$\because\ AM\geq GM\Rightarrow\frac{5\tan^2x+125\cot^2x}{2}\geqslant25$$

$\Rightarrow y \geq 57$

4. **Ans (0)**

$$\int\frac{x+(\cos^{-1}3x)^2}{\sqrt{1-9x^2}}dx$$

$$=\frac{-1}{9}\left(\sqrt{1-9x^2}+(\cos^{-1}3x)^3\right)+C$$

$\therefore\ \alpha = -9, \beta = 3$

$\Rightarrow\ (\alpha + 3\beta) = 0$

5. **Ans (8)**

$$\int\frac{(4+3x)dx}{(x^3-2x-4)}$$

using partial fractions

Let $\frac{4+3x}{(x-2)(x^2+2x+2)}=\frac{A}{x-2}+\frac{Bx+C}{x^2+2x+2}$

$\Rightarrow\ 4 + 3x = A(x^2 + 2x + 2) + (Bx + C)(x - 2)$

Put $x = 0, 2, -1$ to get

$A = 1, B = C = -1$

$$\Rightarrow I=\ell n|x-2|-\frac{1}{2}\int\frac{(2x+2)dx}{x^2+2x+2}\Rightarrow I$$

$$=\ell n\left|\frac{x-2}{\sqrt{x^2+2x+2}}\right|+C\Rightarrow k^2+\ell^2=8$$

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6. **Ans (4)**

$\because x \in (0,1)$

$$\therefore \int \frac{|\ell nx|}{\sqrt{1-x}}dx = -\int \frac{\ell nx}{\sqrt{1-x}}dx$$

$$= -\int \underset{I}{\ell nx}\,\underset{I}{(1-x)^{-\frac{1}{2}}}\,dx$$

$$= -\left(-\ell nx\frac{(1-x)^{\frac{1}{2}}}{\frac{1}{2}} + 2\int \frac{(1-x)^{\frac{1}{2}}}{x}dx\right)$$

$$= 2\sqrt{1-x}\ell nx - 4\int \frac{t^2}{(-1+t^2)}dt$$

(put $\sqrt{1-x} = t$)

$$= 2\sqrt{1-x}\ell nx - 4t - 2\ell n\left|\frac{t-1}{t+1}\right| + C$$

$$= 2\sqrt{1-x}\,(\ell nx - 2) - 2\ell n\left|\frac{\sqrt{1-x}-1}{\sqrt{1-x}+1}\right| + C$$

$\Rightarrow f(x) = \ell nx - 2$ and $f'(x) = \frac{1}{x}$

PART-3 : MATHEMATICS

SECTION-IV

1. **Ans (A->S,B->R,C->T,D->Q)**

(A) $\tan42°(\tan16°\tan44°\tan76°)$

$= \tan42°\ \tan48° = 1$

(B) $(1 - \cot18°)(1 - \cot27°)$ is equal to 2

(C) $\prod_{k=1}^{15} \sin\left(\frac{k\pi}{10}\right)$ is equal to 0

$$\sin\frac{\pi}{10}.\sin\frac{2\pi}{10}\ldots\underbrace{\sin\frac{10\pi}{10}}_{0}.\sin\frac{11\pi}{10}\ldots\sin\frac{15\pi}{10}$$

(D) $\cos^2 65° + \cos^2 55° + \cos65°\cos55°$

$1 - \sin^2 65° + \cos^2 55° + \frac{2\cos65°\cos55°}{2}$

$1 + \cos120°.\cos10° + \frac{1}{2}(\cos120° + \cos10°)$

$1 - \frac{1}{4} = \frac{3}{4}$

2. **Ans (A->P,B->S,C->R,D->Q)**

(A) $\int \frac{dx}{x+2} = \ell n|x+2| + \lambda$

$g(x) = \ell n(x+2)$, $g(0) = \ell n2 \Rightarrow g(-1) = 0$

(B) $I = \int (\sec x)^{9/5}(\operatorname{cosec} x)^{1/5}dx$

$= \int \frac{\sec^2 x\ dx}{(\tan x)^{1/5}}$

Let $\tan x = t^5$

$\sec^2 x\ dx = 5\,t^4\,dt$

$\Rightarrow I = 5\int \frac{t^4\,dt}{t} = \frac{5\,t^4}{4} = \frac{5}{4}(\tan x)^{4/5} + C$

$\Rightarrow km = 1$

(C) $\int \frac{dx}{\cot^2 x - 1} = \int \frac{\sin^2 x dx}{\cos 2x}$

$$2I = \int \frac{(1-\cos2x)dx}{\cos2x} = \int \sec2x dx - \int dx$$

$2I = \frac{1}{2}\ell n|\sec2x + \tan2x| - x$

$\Rightarrow I = \frac{1}{4}\ell n|\sec2x + \tan2x| - \frac{x}{2} + C$

$\Rightarrow \ell - m = 2$

(D) $I = \int \frac{(x+1)e^x dx}{xe^x\sqrt{1+xe^x}}$

Let $1 + xe^x = t^2$

$\Rightarrow (x+1)e^x dx = 2tdt$

$\Rightarrow I = 2\int \frac{tdt}{(t^2-1)t}$

$= \ell n\left|\frac{t-1}{t+1}\right| + C = \ell n\left|\frac{\sqrt{1+xe^x}-1}{\sqrt{1+xe^x}+1}\right| + C$

$\Rightarrow g^2(x) = 1 + xe^x$

$\Rightarrow g^2(1) = 1 + e$

$\Rightarrow g^2(1) = 3.71828 \Rightarrow [g^2(1)] = 3$

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